Problem Set 2 - IC104

Questions

Essential Mathematics for Economic Analysis, 4th edition

Page 118. Question 9

Suppose that all prices rise at the same proportional rate in a country whose inflation rate is 19% per year. For an item that currently costs P_0, use the implied formula:

$P(t) = P_0(1.19)^t$

for the price after t years in order to predict the prices of:

(a) A 20 kg bag of corn, presently costing \$16, after 5 years.

(b) A \$4.40 can of coffee after 10 years.

(c) A \$250,000 house after 4 years.

Page 123. Question 2

Solve the following equations for x:

(a) $3^x = 8$; (b) $ln(x) = 3$; (c) $ln(x^2 - 4x + 5) = 0$

(d) $ln[x(x-2)] = 0$; (e) $\frac{x \cdot ln(x+3)}{x^2+1} = 0$; (f) $ln(\sqrt{x} - 5) = 0$

Page 162. Question 5 a) b) c) d)

In each case, find the slope of the tangent to the graph of f at the specified point:

(a) $f(x) = 3x + 2$ at (0, 2); (b) $f(x) = x^2 - 1$ at (1,0)

(c) $f(x) = 2 + 3/x$ at (3,3); (d) $f(x) = x^3 - 2x$ at (0,0)

Page 177. Question 3

Find the derivatives of the following:

(a) x^6 ; (b) $3x^{11}$; (c) x^{50} ; (d) $-4x^{-7}$

(e) $\frac{x^{12}}{12}$; (f) $-\frac{2}{x^2}$; (g) $\frac{3}{x^{\frac{1}{3}}}$; (h) $-\frac{2}{x\sqrt{(x)}}$

Page 183. Question 2 b) c)

Differentiate:

(b) $(2x^2-1)(x^4-1)$; c) $(x^5+\frac{1}{x})(x^5+1)$

Page 183. Question 3 d) h)

(d) $\frac{x+1}{x-1}$; h) $\frac{3x-1}{x^2+x+1}$

Page 187. Question 2

Compute the following:

(a) $\frac{dY}{dt}$ when $Y = -3(\mathrm{V}+1)^5$ and $V = \frac{1}{3}t^3$

(b) $\frac{dK}{dt}$ when $K = Al^a$ and $L = bt + c$ (A, a , b and c are positive constants.)

Page 193. Question 1

Compute the second derivatives of:

(a) $y = x^5 - 3x^4 + 2$; (b)$y = \sqrt{x}$; (c) $y = (1+x^2)^{10}$

Page 197. Question 3 (a) (b)

Find the first and second-order derivatives of:

(a) $y = e^{-3x}$; (b)$y = 2e^{x^3}$